Name and formula

Reference code: 01-087-3651

Compound name: Silicon Carbide

Empirical formula: CSi
Chemical formula: SiC

Crystallographic parameters

Crystal system: Cubic
Space group: F-43m
Space group number: 216

a (Å): 4.3679
b (Å): 4.3679
c (Å): 4.3679
Alpha (°): 90.0000
Beta (°): 90.0000
Gamma (°): 90.0000

Volume of cell (10^6 pm^3): 83.33
Z: 4.00

RIR: 3.54

Status, subfiles and quality

Status: Alternate Pattern
Subfiles: Alloy, metal or intermetalic, Ceramic, Ceramic - Semiconductor, Common Phase, Forensic, ICSD Pattern, Inorganic
Quality: Indexed (I)

Comments

ANX: NO
ICSD collection code: 101175
Creation Date: 9/1/2021
Cross-References: ICSD:101175, ICDD:04-002-9070

ANX: NO
Analysis: C1 Si1
Formula from original source: Si C
ICSD Collection Code: 101175
ICSD Keywords: HT-HP synthesis; Microstructure; Vickers hardness; Scanning electron microscopy (SEM); Transmission electron microscopy (TEM); Energy-dispersive X-ray spectroscopy (EDX)
Calculated Pattern Original Remarks: Sample synthesized at 1850 C; there is also a second 2H-AlN phase
Structures: Sphalerite
Article Title: Composition and properties of hot-pressed Si C - Al N solid solutions
Temperature of Data Collection: 293 K
Minor Warning: No e.s.d reported/abstracted on the cell dimension. No R factors reported/abstracted
Wyckoff Sequence: c a
Unit Cell Data Source: Powder Diffraction.

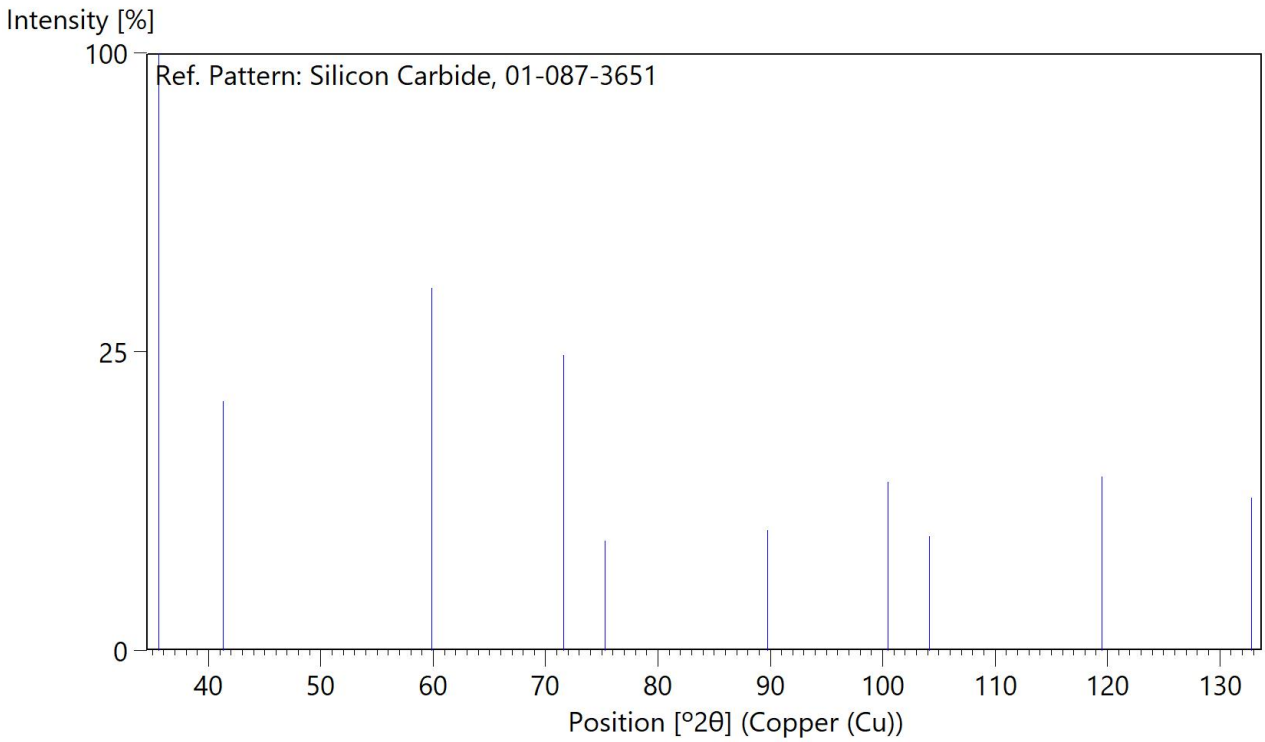
References

Primary reference: Ruh, Robert;Zangvil, Avigdor, *J. Am. Ceram. Soc.*, **65**, 260, (1982)
Structure: Ruh, Robert;Zangvil, Avigdor, *J. Am. Ceram. Soc.*, **65**, 260, (1982)

Peak list

No.	h	k	l	d [Å]	2θ [°]	I [%]
1	1	1	1	2.52181	35.571	100.0
2	2	0	0	2.18395	41.306	17.5
3	2	2	0	1.54429	59.842	36.9
4	3	1	1	1.31697	71.592	24.5
5	2	2	2	1.26090	75.311	3.4
6	4	0	0	1.09197	89.727	4.1
7	3	3	1	1.00206	100.477	8.0
8	4	2	0	0.97669	104.125	3.7
9	4	2	2	0.89159	119.528	8.5
10	5	1	1	0.84060	132.801	6.6

Structure

No.	Name	Element	X	Y	Z	Biso	sof	Wyck.
1	C1	C	0.00000	0.00000	0.00000	0.5000	1.0000	4a
2	SI1	Si	0.25000	0.25000	0.25000	0.5000	1.0000	4c

Stick Pattern